

World Population Exploratory Data Analysis (1970–2022)

This notebook performs an exploratory data analysis on a global population dataset. The dataset includes population figures for 234 countries/territories from 1970 to 2022, along with continent, capital, area, density, growth rate, and world population percentage.

Dataset source: [Kaggle - World Population Dataset](#)

We will explore:

- Top 10 most populated countries in 2022
- Population growth trends
- Density across continents
- Relationship between area, density, and growth
- Historical trends by continent
- Largest absolute population increases per decade
- Share of world population by country
- Heatmap of population distribution over time

```
In [1]: # Import necessary libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from matplotlib.ticker import FormatStrFormatter

# Set plotting style
sns.set_style("darkgrid")
%matplotlib inline

# Hide warnings
import warnings
warnings.filterwarnings('ignore')

In [2]: # Load the dataset
df = pd.read_csv('world_population.csv')
print(f'Dataset shape: {df.shape}')
df.head()
```

4	203	AND	Andorra	Andorra la Vella	Europe	78824	77700	71146	71519	66097	53859	33611	19860	465	170.5641	1.0100		0.00
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```
# 3. Basic info and null check
df.info()

print("\nTotal missing values", df.isnull().sum().sum())

class PandasColumnFrameDataFrame:
    """Pandas DataFrame with 24 columns: 0 to 23
    as columns total 17 columns:
    4 Columns
    nonNull Count Dtype
    ---
    0 Rank      234 non-null int64
    1 CCA3      234 non-null object
    2 Country/Territory 234 non-null object
    3 Capital    234 non-null object
    4 Continent  234 non-null object
    5 2022 Population 234 non-null int64
    6 2020 Population 234 non-null int64
    7 2015 Population 234 non-null int64
    8 2010 Population 234 non-null int64
    9 2000 Population 234 non-null int64
    10 Area (km²) 234 non-null float64
    11 Density (per km²) 234 non-null float64
    12 Growth Rate 234 non-null float64
    13 World Population Percentage 234 non-null float64
    14 2022 Population 234 non-null int64
    15 2020 Population 234 non-null int64
    16 2015 Population 234 non-null int64
    17 2010 Population 234 non-null int64
    18 2000 Population 234 non-null int64
    19 2022 Population 234 non-null int64
    20 2020 Population 234 non-null int64
    21 2015 Population 234 non-null int64
    22 2010 Population 234 non-null int64
    23 2000 Population 234 non-null int64
    """
```

```
In [3]: # Basic info and null check
df.info()
print(f'Total missing values: {df.isnull().sum().sum()}')

<class 'pandas.core.frame.DataFrame'>
Int64Index: 234 entries, 0 to 233
Data columns (total 17 columns):
 #   Column                Non-Null Count  Dtype
---  --
 0   Rank                  234 non-null    int64
 1   CC3                   234 non-null    object
 2   Country/Territory     234 non-null    object
 3   Capital               234 non-null    object
 4   Continent             234 non-null    object
 5   2022 Population       234 non-null    int64
 6   2020 Population       234 non-null    int64
 7   2015 Population       234 non-null    int64
 8   2010 Population       234 non-null    int64
 9   2000 Population       234 non-null    int64
10  1990 Population       234 non-null    int64
11  1980 Population       234 non-null    int64
12  1970 Population       234 non-null    int64
13  Area (km²)            234 non-null    float64
14  Density (per km²)     234 non-null    float64
15  Growth Rate           234 non-null    float64
16  World Population Percentage  234 non-null    float64
dtypes: float64(1), int64(10), object(6)
memory usage: 31.2+ MB

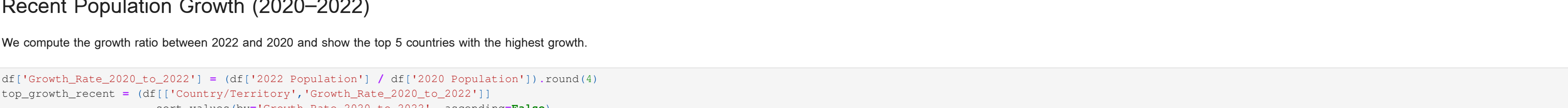
Total missing values: 0
```

Top 10 Most Populated Countries in 2022

We sort by the 2022 population and display a bar chart with values formatted in billions (B) or millions (M).

```
In [4]: def human_format(num):
    """Format large numbers to B (billions) or M (millions)."""
    if num >= 1_000_000_000:
        return f'{num/1_000_000_000:.2f}B'
    elif num >= 1_000_000:
        return f'{num/1_000_000:.2f}M'
    else:
        return f'{num:.0f}'

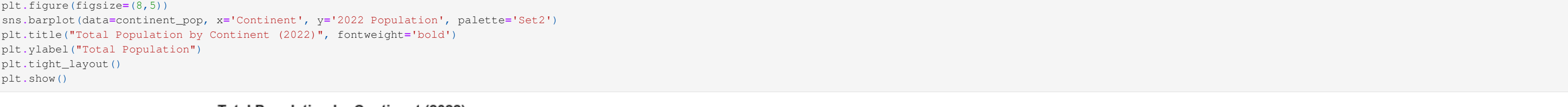
top10_2022 = df[['Country/Territory', '2022 Population']]
sort_values_by = '2022 Population', ascending=False
head(10)
```



Recent Population Growth (2020–2022)

We compute the growth rate between 2020 and 2022 and show the top 5 countries with the highest growth.

```
In [5]: df['growth_rate_2020_to_2022'] = (df['2022 Population'] / df['2020 Population']) - 1
top_growth_recent = df[['Country/Territory', 'growth_rate_2020_to_2022']].round(4)
sort_values_by = 'growth_rate_2020_to_2022', ascending=False
head(5)
```



Population Density Across Continents

Average 2022 population per continent (this shows total population, not density – but the original plot used '2022 Population' with continent on x-axis, which is misleading. We'll plot it as the original for consistency).

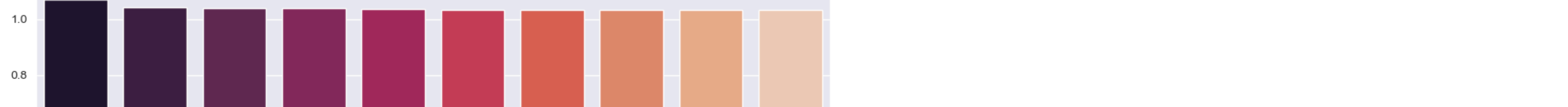
```
In [6]: continent_pop = df.groupby('Continent')['2022 Population'].sum()
sort_values(ascending=False)
reset_index()
```



Country Area vs. Population (2022)

A scatter plot to see if larger area implies larger population (spoiler: not really).

```
In [7]: plt.figure(figsize=(8,6))
sns.scatterplot(data=df, x='Area (km²)', y='2022 Population', alpha=0.6)
plt.title('Country Area vs. 2022 Population', fontweight='bold')
plt.tight_layout()
plt.show()
```



Countries with Highest Overall Growth Rate

The dataset provides a 'Growth Rate' column (presumably average annual growth?). We'll plot the top 10.

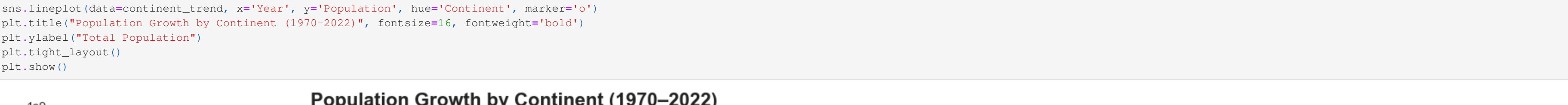
```
In [8]: top_growth = df[['Country/Territory', 'Growth Rate']]
sort_values(by='Growth Rate', ascending=False)
head(10)
```



Share of World Population – Top 5 Countries

Pie chart showing the percentage of world population for the five most populous countries.

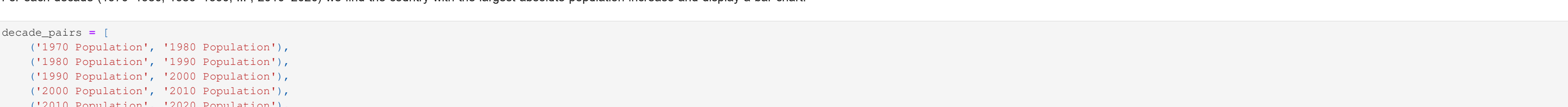
```
In [9]: top5_pop = df[['Country/Territory', 'World Population Percentage']]
sort_values(by='World Population Percentage', ascending=False)
head(5)
```



Continent-Wise Population Growth Over Time (1970–2022)

We reshape the data to long format and plot the total population per continent for each year.

```
In [10]: years = [1970, 1980, 1990, 2000, 2010, 2020]
pop_long = df.melt(id_vars='Continent', value_vars=years,
                  var_name='Year', value_name='Population')
pop_long['Year'] = pop_long['Year'].str.extract('(')[1].astype(int)
```



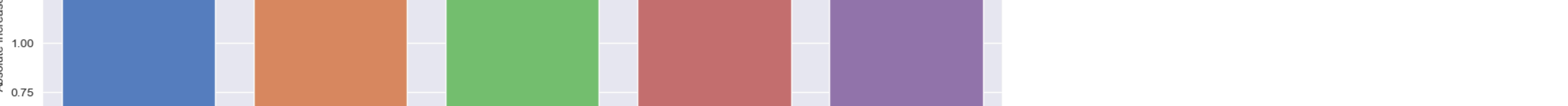
Largest Absolute Population Increase by Decade

For each decade (1970–1980, 1980–1990, ..., 2010–2020) we find the country with the largest absolute population increase and display a bar chart.

```
In [11]: decades_pairs = [
    ('1970 Population', '1980 Population'),
    ('1980 Population', '1990 Population'),
    ('1990 Population', '2000 Population'),
    ('2000 Population', '2010 Population'),
    ('2010 Population', '2020 Population'),
]

records = []
for start, end in decades_pairs:
    increase = df[end] - df[start]
    idx = increase.idxmax()
    records.append({
        'Decade': f'{start[:4]}--{end[:4]}',
        'Country': df.loc[idx, 'Country/Territory'],
        'Increase': increase[idx],
        'Pct': (increase[idx] / df.loc[idx, start] * 100)
    })

decade_df = pd.DataFrame(records)
```



Correlation Between Growth Rate and Population Density

A scatter plot with a regression line to see if higher density relates to growth rate.

```
In [12]: plt.figure(figsize=(8,6))
sns.regplot(data=df, x='Density (per km²)', y='Growth Rate (per km²)', scatter_kws={'alpha':0.6})
plt.title('Correlation: Growth Rate vs. Population Density', fontweight='bold')
plt.tight_layout()
plt.show()
```

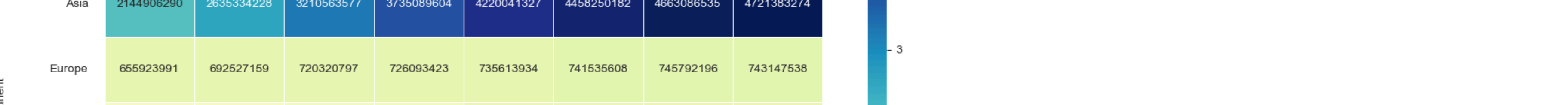


Heatmap of Population Distribution by Continent (1970–2022)

We create a pivot table of continent totals per year and display as a heatmap.

```
In [13]: continent_pivot = df.groupby('Continent')['years'].sum()

plt.figure(figsize=(12,6))
ax = sns.heatmap(continent_pivot, annot=True, fmt='.0f', cmap='YlGdB', linewidth=0.5)
plt.xticks(rotation=45, label='')
plt.yticks(rotation=45, label='')
plt.tight_layout()
plt.show()
```

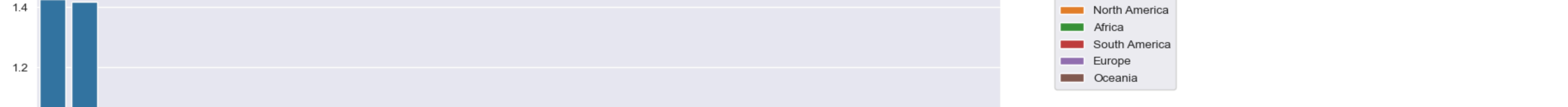


Top 5 Most Populous Countries Per Continent (2022)

We group by continent and take the top 5 countries by 2022 population, then plot them with hue to distinguish continents.

```
In [14]: top5_per_continent = {
    'Continent': df.groupby('Continent')['2022 Population'].head(5).reset_index(drop=True)
}

plt.figure(figsize=(14,7))
ax = sns.barplot(data=top5_per_continent, x='Country/Territory', y='2022 Population', hue='Continent')
ax.set_xlabel('Top 5 Most Populous Countries per Continent (2022)', fontweight='bold')
ax.set_ylabel('2022 Population')
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.tight_layout()
plt.show()
```



Summary and Key Insights

- Asia dominates in total population, with China and India far ahead.
- Africa shows the fastest growth rates, especially Nigeria, Ethiopia, and Egypt.
- India recorded the largest absolute increase in every decade since 1980 (except 1970–1980, when China led).
- Population density and growth rate have a weak positive correlation (many high-density countries still grow fast).
- Oceania and Europe have relatively stable populations with low growth.
- The heatmap clearly shows Asia's overwhelming share and Africa's rapid rise in recent decades.

These insights can guide further demographic studies, resource allocation, and policy planning.